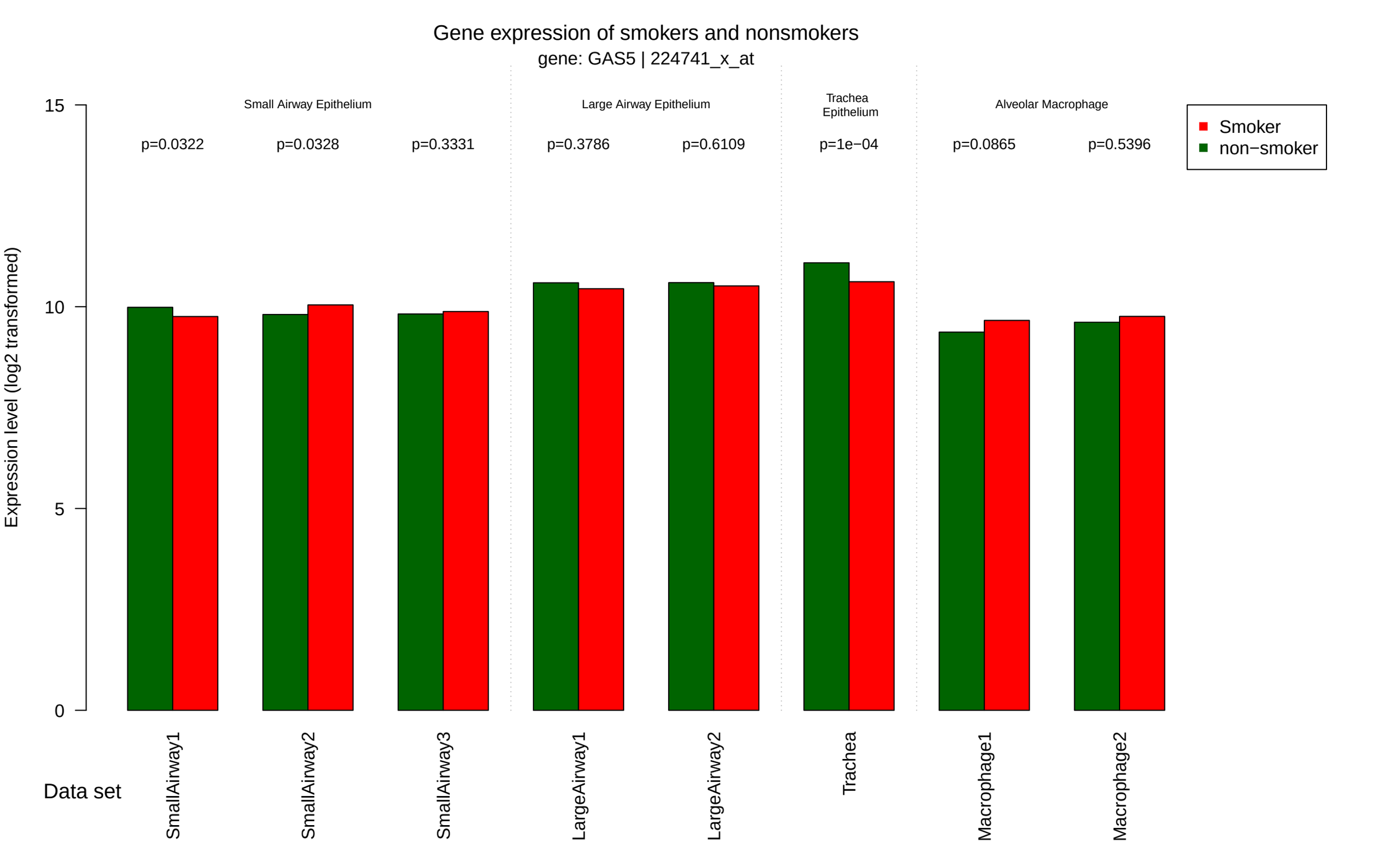
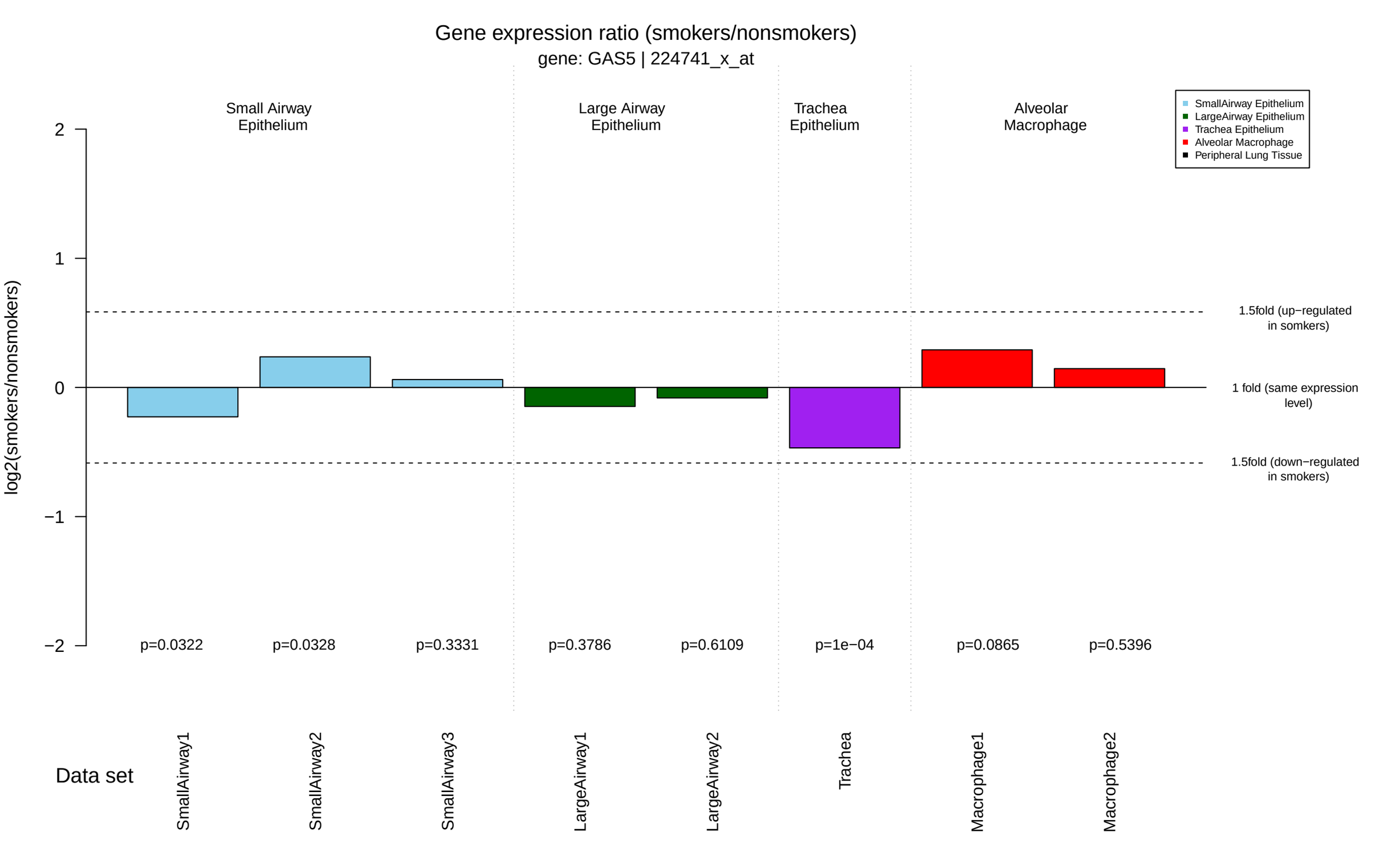
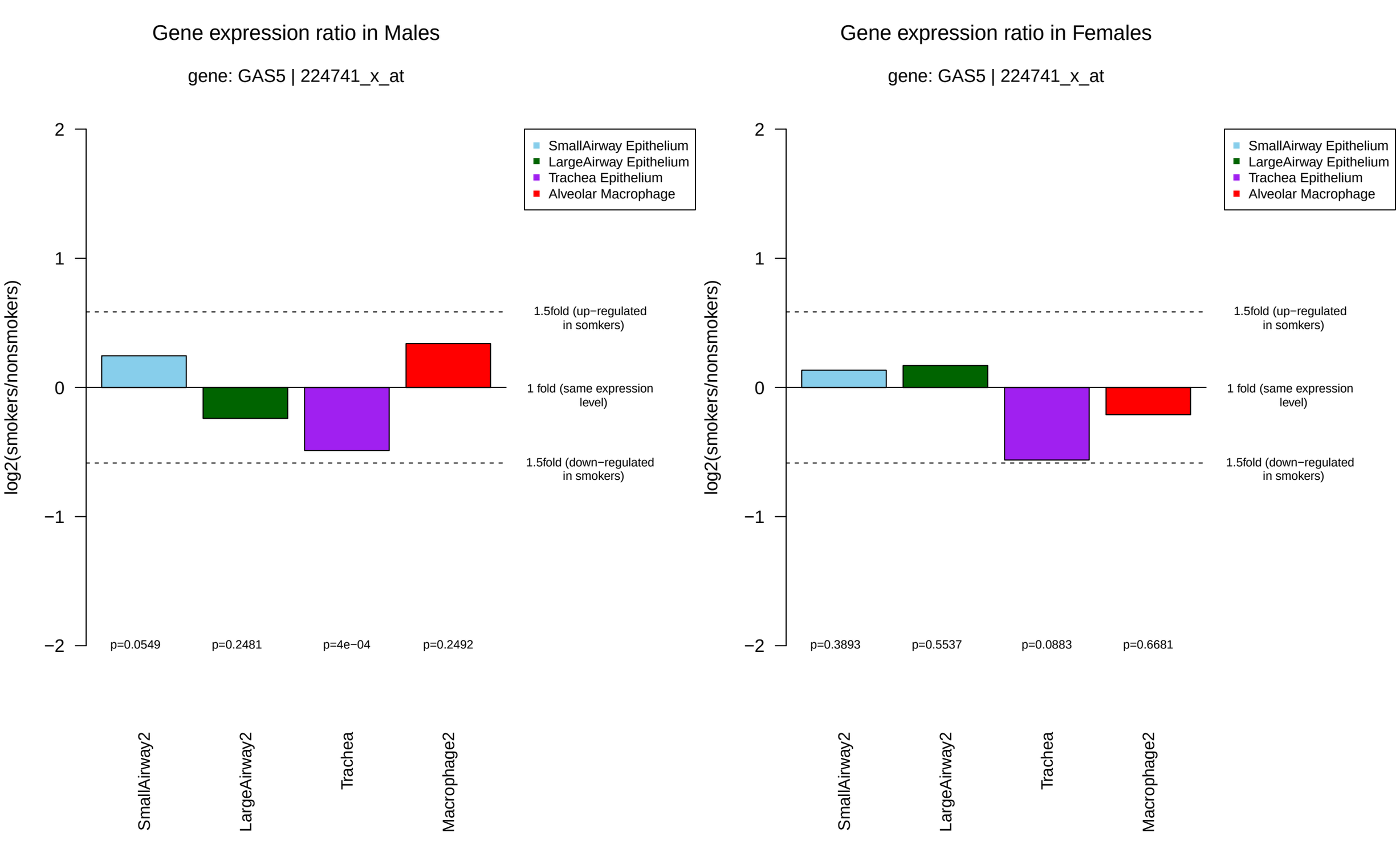
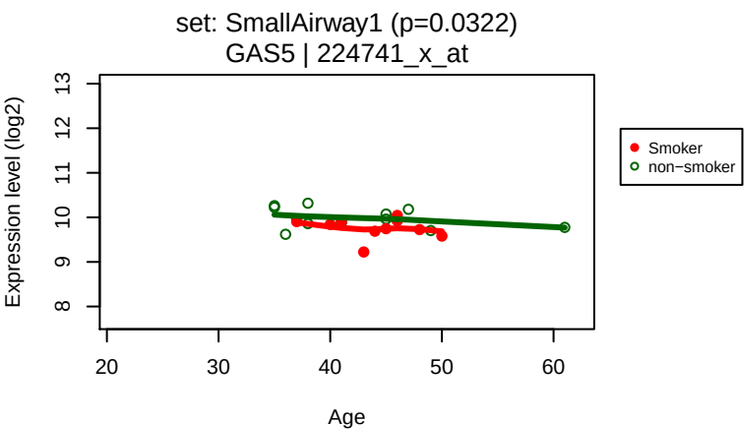
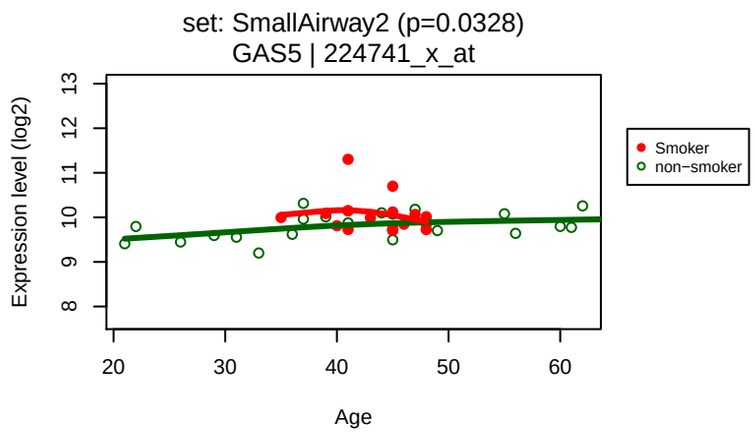
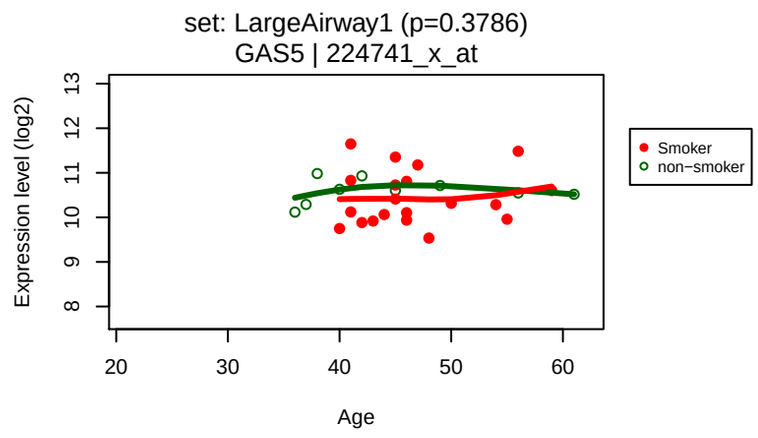

Gene expression of smokers and nonsmokers
gene: GAS5 | 224741_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0322
p=0.0328
p=0.3331
p=0.3786
p=0.6109
p=1e−04
p=0.0865
p=0.5396
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

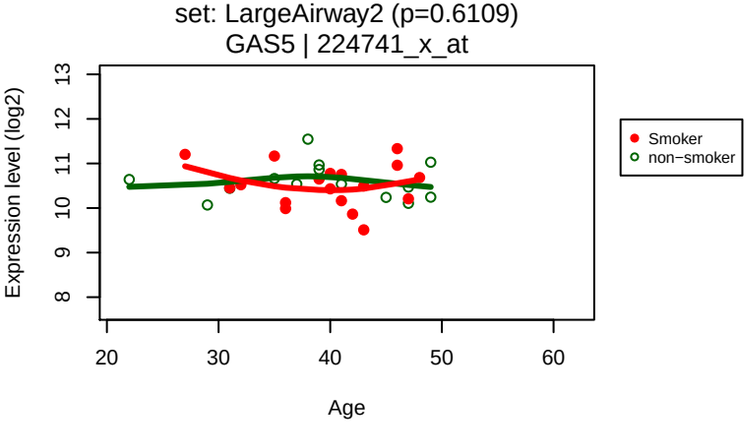

Gene expression ratio (smokers/nonsmokers)
gene: GAS5 | 224741_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0322
p=0.0328
p=0.3331
p=0.3786
p=0.6109
p=1e−04
p=0.0865
p=0.5396
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

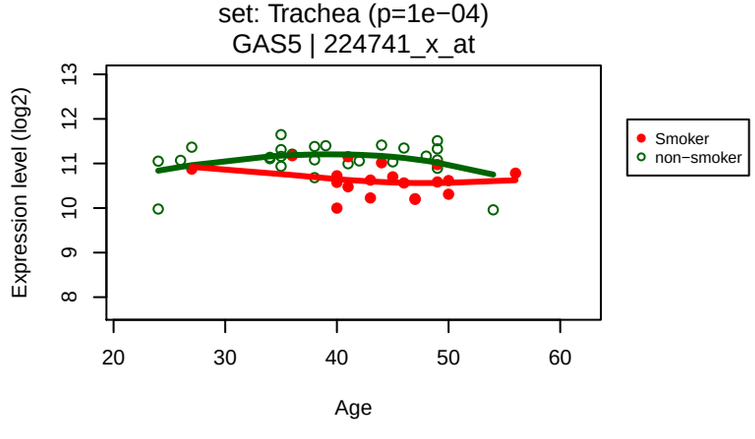

Gene expression ratio in Males
gene: GAS5 | 224741_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0549
p=0.2481
p=4e−04
p=0.2492
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAS5 | 224741_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3893
p=0.5537
p=0.0883
p=0.6681
SmallAirway2
LargeAirway2
Trachea
Macrophage2

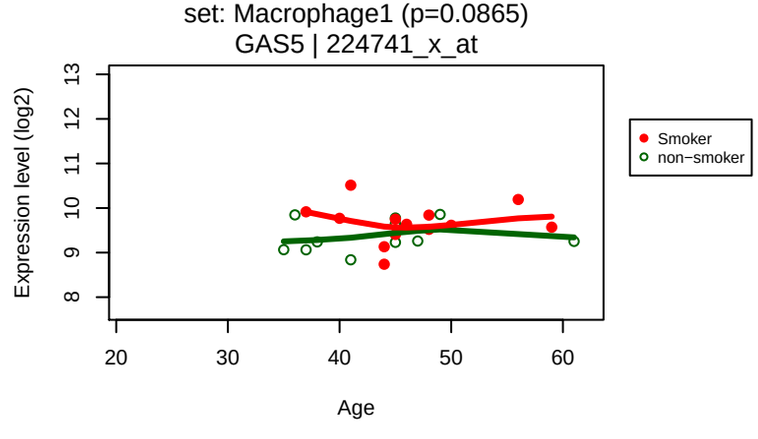

set: SmallAirway1 (p=0.0322)
GAS5 | 224741_x_at
Expression level (log2)
Age
Smoker
non−smoker

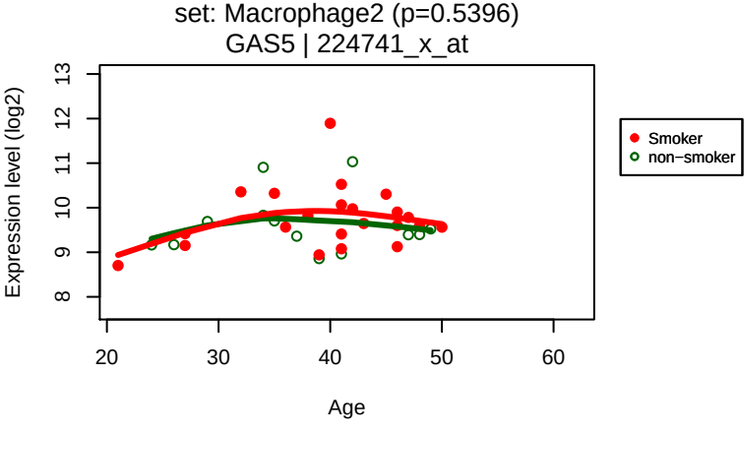

set: SmallAirway2 (p=0.0328)
GAS5 | 224741_x_at
Expression level (log2)
Age
Smoker
non−smoker

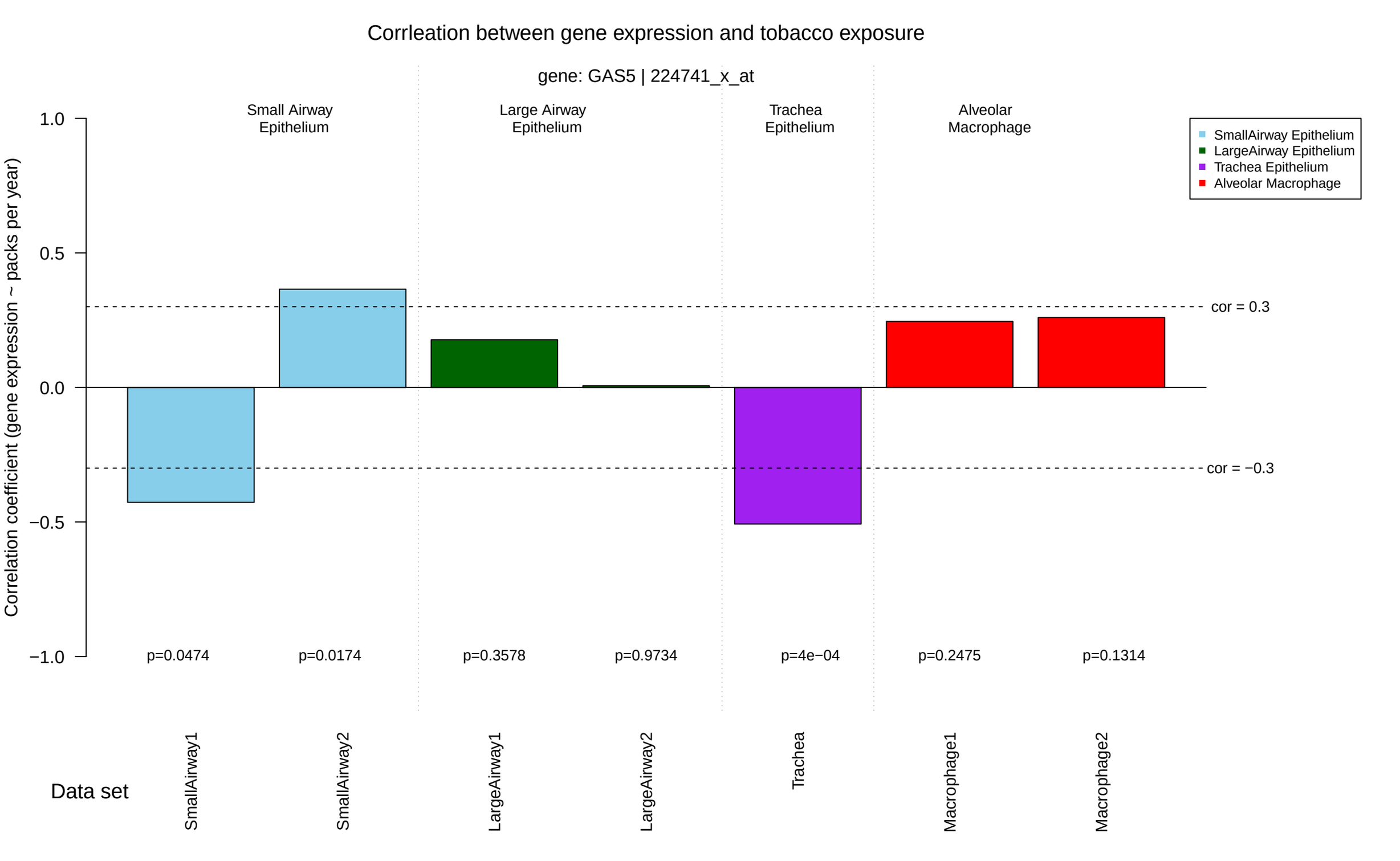

set: LargeAirway1 (p=0.3786)
GAS5 | 224741_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6109)
GAS5 | 224741_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
GAS5 | 224741_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0865)
GAS5 | 224741_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5396)
GAS5 | 224741_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAS5 | 224741_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0474
p=0.0174
p=0.3578
p=0.9734
p=4e−04
p=0.2475
p=0.1314
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2